

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 SoPS Lecture 6
 - Battleship
 - Group assignment: Stealthy fleets
 - Wrapping up

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- Battleship
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Chomp Review

Last time

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- we explored Battleship

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- we looked at basic probability

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- we explored Battleship
- we looked at basic probability
- we learned how to apply probability to Battleship

Battleship with Radar

Radar Systems:

In Battleship with Radar, both players have **radar data** giving the occupied spots in each row and column.

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Radar Example:

X					X				
X		X	X		X				
X					X				
		X			X				
		X			X				
		X							
		X			X	X	X		

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x					x				
x		x	x		x				
x					x				
		x			x				
		x			x				
		x							
		x			x	x	x		

Rows: 2, 4, 2, 2, 2, 1, 4, 0, 0, 0

Battleship with Radar

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Radar Example:

x					x				
x		x	x		x				
x					x				
		x			x				
		x			x				
		x							
		x			x	x	x		

Rows: 2, 4, 2, 2, 2, 1, 4, 0, 0, 0 Cols: 3, 0, 5, 1, 0, 6, 1, 1, 0, 0

Battleship with Radar

Activity (30 minutes):

Play Battleship with Radar

- set up the boards normally
- then tell your opponent the radar data
- then play Battleship normally

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Challenge:

- record initial setups
- what makes a good setup?

Radar: Debrief

Radar: Debrief

Submit pictures of good/bad setups:



Radar: Debrief

Submit pictures of good/bad setups:



Is it possible to guess the entire fleet from radar?

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Inverse problems

Vocabulary

The problem of determining a Battleship fleet from its X-ray data is an example of an **inverse problem**. More generally, inverse problems arise when we try to reconstruct something from a set of observations.

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Natural questions:

Given some X-ray data, we can ask:

Inverse problems

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Given some X-ray data, we can ask:

- does a fleet with that data exist

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Given some X-ray data, we can ask:

- does a fleet with that data exist
- if so, how can we reconstruct it?

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Vocabulary

The problem of determining a Battleship fleet from its X-ray data is an example of an **inverse problem**. More generally, inverse problems arise when we try to reconstruct something from a set of observations.

Natural questions:

Given some X-ray data, we can ask:

- does a fleet with that data exist
- if so, how can we reconstruct it?
- is there only one possible fleet?

Group assignments

Group creation

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- create groups of 3 – 4 people

Group assignments

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- create groups of 3 – 4 people
- take a minute to say hello to your members

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- create groups of 3 – 4 people
- take a minute to say hello to your members
- choose a group name

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Submitting assignments

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- create a Google doc

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Submitting assignments

- create a Google doc
- carefully type up answers / insert images

Group assignments

Group creation

- create groups of 3 – 4 people
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Submitting assignments

- create a Google doc
- carefully type up answers / insert images
- submit on Canvas as a single PDF

Group assignments

Group creation

- create groups of 3 – 4 people
- take a minute to say hello to your members
- choose a group name
- self-sign up on Canvas

Submitting assignments

- create a Google doc
- carefully type up answers / insert images
- submit on Canvas as a single PDF
- only one member needs to submit

Existence

Problem 1

Existence

Problem 1

- (a) Find a Battleship fleet with X-ray data
Rows: 4, 4, 4, 4, 1, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0

Existence

Problem 1

- (a) Find a Battleship fleet with X-ray data
Rows: 4, 4, 4, 4, 1, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0
- (b) Find a Battleship fleet with X-ray data
Rows: 0, 2, 5, 1, 4, 2, 1, 1, 1, 0 Cols: 0, 1, 2, 1, 2, 5, 1, 1, 1, 3

Non-existence

Problem 2

Non-existence

Problem 2

- (a) Explain why there is no fleet with the *X-ray* data Rows:
4, 2, 4, 3, 3, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0

Non-existence

Problem 2

- (a) Explain why there is no fleet with the *X*-ray data Rows: 4, 2, 4, 3, 3, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0
- (b) Explain why there is no fleet with the *X*-ray data Rows: 0, 2, 2, 5, 4, 4, 0, 0, 0, 0 Cols: 1, 2, 2, 3, 2, 2, 0, 2, 1, 2

Non-uniqueness

Problem 3

Non-uniqueness

Problem 3

- (a) Find an example of a fleet whose position is uniquely determined by its radar data.

Non-uniqueness

Problem 3

- (a) Find an example of a fleet whose position is uniquely determined by its radar data.
- (b) Find an example of a fleet whose position is “stealthy”: whose radar data actually describes several different potential fleet positions. Show the other fleets.

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Existence and uniqueness

Many theorems in mathematics focus on the **existence** of something. Given an equation, does a solution exist? The natural follow-up is **uniqueness**. Is there a unique solution to the problem or are there many?

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Question 1

Can you think of a real-world example of an inverse problem?

Discussion

Existence and uniqueness

Many theorems in mathematics focus on the **existence** of something. Given an equation, does a solution exist? The natural follow-up is **uniqueness**. Is there a unique solution to the problem or are there many?

Question 1

Can you think of a real-world example of an inverse problem?

Question 2

What are some necessary conditions on radar data for it to come from a Battleship fleet?

Question of the Day

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Puzzle

Determine the number of ways to put a 2×1 destroyer and a 3×1 submarine on the 10×10 Battleship board at the same time.

Final thoughts

Reflection

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- What situations in the real world can you think of where existence and uniqueness questions arise?

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Reflection

- What situations in the real world can you think of where existence and uniqueness questions arise?
- Can you think of any other interesting ways to modify the game Battleship?

Submit your responses by **Tonight!**